A correct calculation can produce a harmful decision

- Arithmetic checks whether a rule was followed
- Evaluation checks whether the rule serves its purpose
- The relevant people may disagree about success



AI conceptual illustration · not an actual company, venue or measured outcome

Replacing and supporting human judgement

> Automation can apply a rule consistently at scale

> Human reviewers can interpret context and exceptions

> A reviewer needs authority, time and useful information

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 86–87.

Bias can enter at several points

- Data may omit or misrepresent a group
- A proxy may measure something other than the intended quality
- An objective or threshold may distribute errors unevenly

Who is represented?	What proxy or label?
Which objective?	Whose results differ?

Bias can enter at several points. Compare effects for relevant groups instead of relying only on a whole-dataset average.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 87.

APPLY · 4 · TEXTBOOK PP. 87

Missing examples can change who succeeds

- A training set contains 90 examples from Group A and 10 from B
- A model’s average result may hide a weak result for B
- Check performance for the people affected



Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 87.

APPLY · 5 · TEXTBOOK PP. 87

A proxy can misrepresent the intended quality

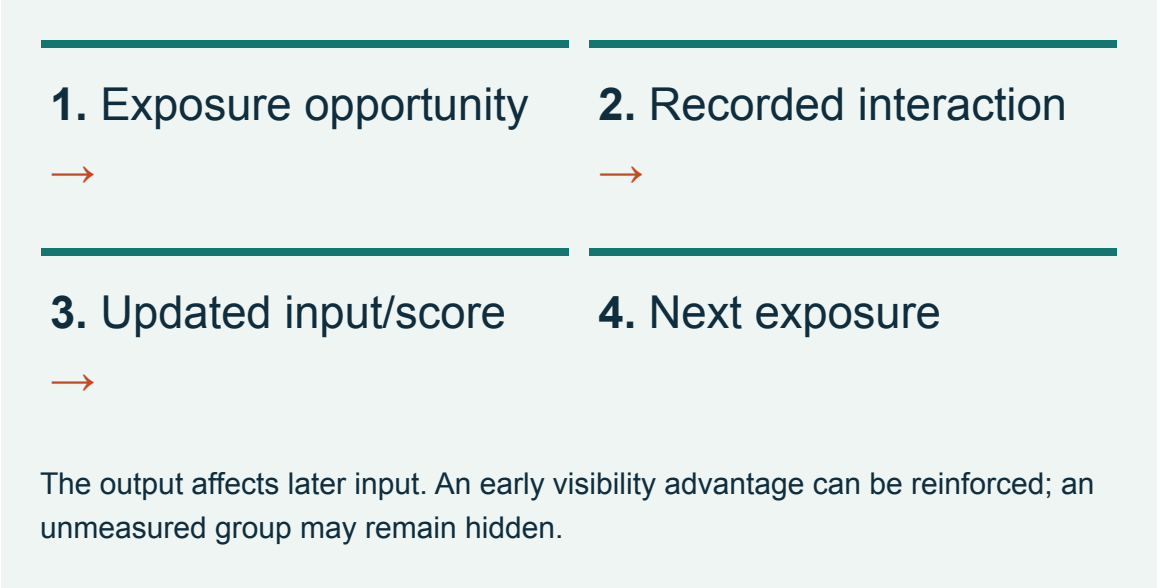
> Desired quality: reliable access to a task

> Measured signal: previous completion count

> Low completion may reflect limited opportunity

A feedback loop can reinforce an early advantage

- Exposure gives an item opportunities for interaction
- Interactions become input for the next ranking
- A low-exposure item may remain poorly measured



The output affects later input. An early visibility advantage can be reinforced; an unmeasured group may remain hidden.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 87–89.

Human judgement can erode through over-reliance

> A score may seem authoritative because it is numerical

> Repeated acceptance can reduce independent checking

> A challenge route helps expose errors

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 86–88.

APPLY · 8 · TEXTBOOK PP. 88

A black box limits explanation

> Inputs and outputs may be visible

> The path from input to result may remain unclear

> Secrecy and model complexity can create different barriers

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 88.

APPLY · 9 · TEXTBOOK PP. 88

Transparency makes parts of the decision understandable

- Explain the objective and significant inputs
- Show how a person can correct inaccurate data
- Provide an explanation suited to the audience

Rule is disclosed	Individual result explained	Limitations made clear

Visibility can help understanding; it does not by itself assign responsibility or provide remedy.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 88.

Accountability assigns responsibility and consequences

- Name who must explain the decision
- Give someone authority to investigate and repair it
- Check whether the repair improves the relevant outcome

1. Responsible actor →

2. Challenge route →

3. Review with authority →

4. Repair or consequence

A route must be usable and capable of affecting a decision; merely adding a human name is insufficient.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 88.

Real case A: a provider explains recommendations

> TikTok describes interaction and video-information signals

> The account presents personalization as a benefit

> Its 2020 description does not reveal today’s exact weights

TikTok, 18 June 2020, provider account. https://newsroom.tiktok.com/en-us/how-tiktok-recommends-videos-for-you Textbook connection: pp. 86–89.

Real case C: a regulator examines design risks

> The European Commission issued preliminary findings in February 2026

> The assessment concerns several interacting design features

> The source states that the investigation’s outcome remains open

European Commission, 6 February 2026, preliminary findings. https://digital-strategy.ec.europa.eu/en/news/commission-preliminarily-finds-tiktoks-addictive-design-breach-digital-services-act Textbook connection: pp. 86–89.

Observation, explanation and intention

> Observation: a measured pattern in the results

> Mechanism: an account of how a rule could produce it

> Intention: a claim about an actor’s purpose

1. Recorded observation →

2. Proposed mechanism →

3. Contextual consequence →

4. Qualified judgement

State which links the source establishes and which are your inferences. Give a condition that could change the judgement.

Bomfim et al., Digital Society for the IB Diploma, Hodder, 2022. Printed pages cited per slide. Original paraphrase. Textbook connection: pp. 87–89.

A conditional judgement weighs an intervention

> Criterion: fair access to visibility

> Proposal: reserve some exposure for less-seen items

> Trade-off: discovery may displace predicted relevance

Paper 1 and Paper 2 use the same knowledge differently

> Paper 1: apply knowledge to a specified situation

> Paper 2: use and compare claims from supplied sources

> Both require precise mechanisms and supported reasoning

IB Digital Society guide, local April 2026 revision, sections 3.1H, 3.2A–E and 4.2A–D. Assessment authority. Textbook connection: pp. 89, 91.

Chapter 3.2: delayed explanation check

> Explain one mechanism without notes

> Apply it to a changed context

> Return to the explanation you could not yet justify